

◦ Aim:

- To verify the Nernst equation and determine the standard electrode potential of copper.

◦ Apparatus:

- Potentiometer, calomel electrode, copper electrode, copper sulphate, distilled water, volumetric flasks, graduated pipette, beakers.

◦ Theory:

- Electrode potential values of any metal are directly linked to the reactivity of the metal. Hence, whenever a new metal is found, it is important to find its electrode potential in order to ascertain its relative reactivity among the metals.
- Every cell is composed of two electrodes. Thus in order to find the potential of a particular electrode under consideration, we combine it with another electrode whose potential is known (eg, calomel electrode). The EMF (electromotive force) can be determined by combining these two electrodes using a potentiometer.
- Potentiometry deals with the measurement of difference in potential between two electrodes which have been combined to form electrochemical cells. In potentiometer, the principle generally used is the Poggendorff compensation method. This consists in balancing the unknown EMF against a known one in a potentiometric circuit. In other words, we measure the external potential difference required to stop the passage of current in a cell.

- The EMF is related to the electrode potential of any element by:-
 $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$ ($E_{\text{cell}} = \text{EMF}$, $E = \text{electrode potential}$).
 - In principle, all standard electrodes are to be measured using a primary standard electrode system such as the hydrogen electrode. But at the same time we know it's difficult to build a primary hydrogen electrode in every laboratory and hence certain secondary electrode systems have been developed such as the calomel electrode.
 - The standard electrode potential (E°) is related to the electrode potential by the equation: $E = E^\circ - \frac{2.303RT}{nF} \log \left(\frac{[\text{Product (m.x.r)}]}{[\text{Reactant (m.x.r)}]} \right)$.
- The above equation is called the Nernst equation using which the standard electrode potential can be found.

o Procedure:

- A stock solution of 0.5M CuSO_4 solution is given. Using the solutions, prepare 0.1, 0.01, 0.05, 0.005M solutions using volumetric flask with distilled water as the solvent.
- Place the electrodes in a beaker and pour the most dilute solution (0.005M) first in the case.
- Connect the electrodes to the potentiometer by following color codes. Put on the instrument. Allow the stabilization time of 10 minutes with proper mixing of the solution. Read and note the EMF value in the table.
- Repeat the step for the remaining solutions.
- After all readings, wash the electrodes with distilled water and place the electrodes in an empty beaker. Put off the instrument.

Observations:

Molarity (m)	E_{cell} (V)	Activity Coefficient (γ)	Activity ($a = m \times \gamma$)	$\log(a)$	E_{Cu} (V)	E_{Cu}° (V)
0.1M	0.057	0.15	0.015	-1.823	0.297	0.3507
0.01M	0.044	0.41	4.1×10^{-3}	-2.387	0.264	0.3344
0.05M	0.052	0.21	0.0105	-1.978	0.292	0.3503
0.005M	0.041	0.48	2.4×10^{-3}	-2.619	0.245	0.3222

Calculations:

- $E_{cell} = E_{cathode} - E_{anode} \Rightarrow E_{cathode} = E_{cell} + E_{anode} \Rightarrow E_{Cu} = E_{cell} + E_{calomel}$

i) $E_{Cu} = 0.057 + 0.24 = 0.297V$

ii) $E_{Cu} = 0.044 + 0.24 = 0.284V$

iii) $E_{Cu} = 0.052 + 0.24 = 0.292V$

iv) $E_{Cu} = 0.041 + 0.24 = 0.281V$

- $E_{Cu} = E_{Cu}^{\circ} + 0.0295 \log(a) \Rightarrow E_{Cu}^{\circ} = E_{Cu} - 0.0295 \log(a)$

i) $E_{Cu}^{\circ} = 0.297 + 0.0295(1.823) = 0.3507V$

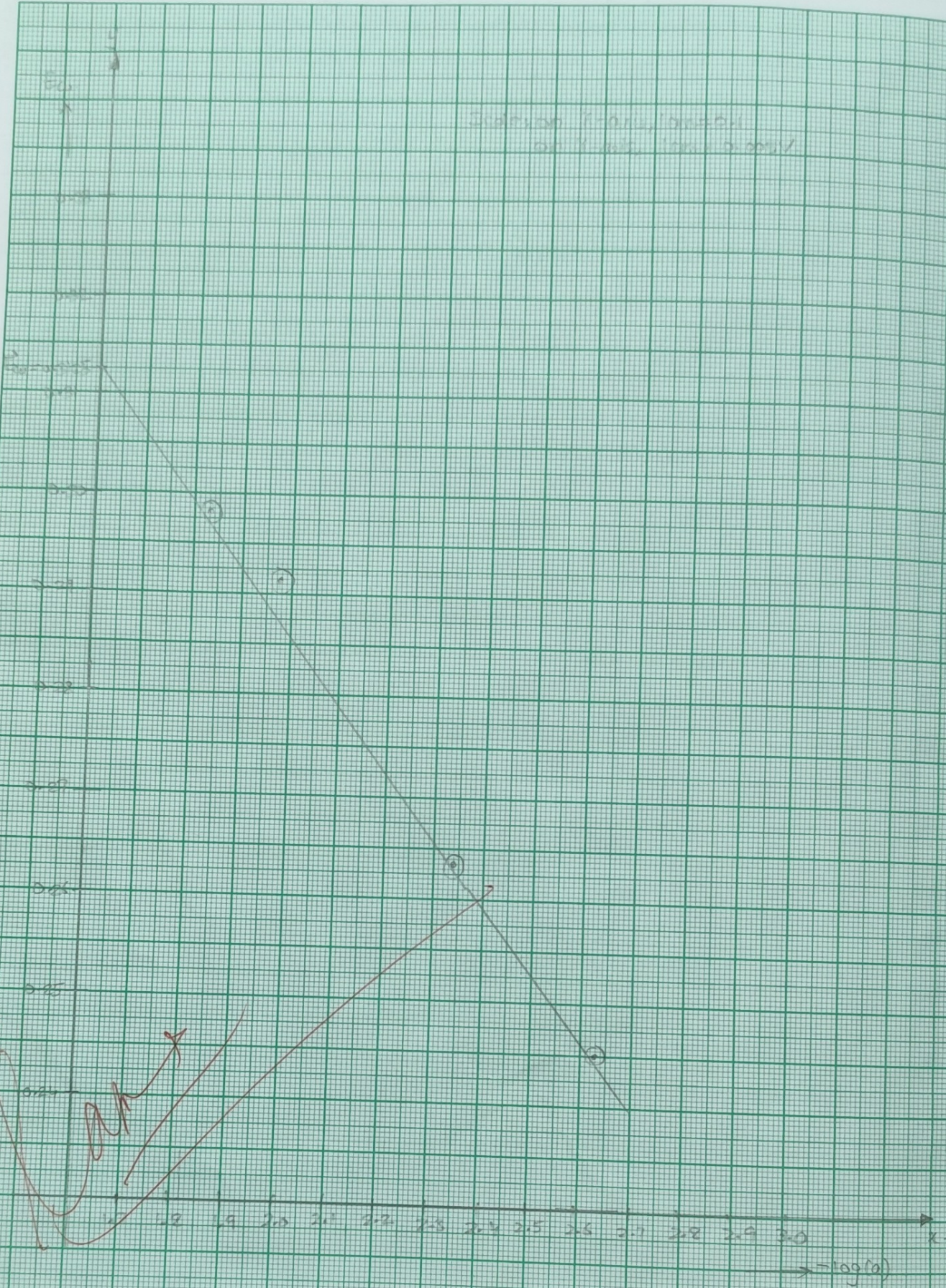
ii) $E_{Cu}^{\circ} = 0.264 + 0.0295(2.387) = 0.3344V$

iii) $E_{Cu}^{\circ} = 0.292 + 0.0295(1.978) = 0.3503V$

iv) $E_{Cu}^{\circ} = 0.245 + 0.0295(2.619) = 0.3222V$

$\therefore (E_{Cu}^{\circ})_{avg} = \frac{0.3507 + 0.3344 + 0.3503 + 0.3222}{4} = 0.3394V$

1. $\log_{10} \frac{1}{a}$ vs $\log_{10} \frac{1}{b}$
 2. $\log_{10} \frac{1}{a}$ vs $\log_{10} \frac{1}{b}$



• Calculations:

- To prepare solutions, $N_1 V_1 = N_2 V_2$.

Where $N_1 = 0.5M$, $V_2 = 50mL$.

- i) For $0.1M$: $0.5 V_1 = (0.1)(50) \Rightarrow V_1 = 10mL$
- ii) For $0.01M$: $0.5 V_1 = (0.01)(50) \Rightarrow V_1 = 1mL$
- iii) For $0.05M$: $0.5 V_1 = (0.05)(50) \Rightarrow V_1 = 5mL$
- iv) For $0.005M$: $0.5 V_1 = (0.005)(50) \Rightarrow V_1 = 0.5mL$

• Precautions:

- Handle the Calomel electrode cautiously.

• Results:

- E°_{Cu} value from calculations = $0.3394V$.
- E°_{Cu} value from graph = $0.3125V$.